

LISTA FYR, Norway

WMO: 014270

Lat: 58.1090N

Lon: 6.5678E

Elev: 46

StdP: 14.67

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	18.5	22.2	7.7	8.2	22.7	12.2	10.3	25.3	41.4	41.9	37.2	41.2	8.7	50	0.778

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	8.2	72.3	62.8	69.1	61.7	66.7	60.7	64.7	69.9	63.0	67.3	61.7	65.4	13.9	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.7	85.3	66.7	61.1	80.7	65.3	59.9	77.0	64.1	29.8	70.1	28.6	67.4	27.6	65.6	72.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 36.9	32.7	29.3	DB	16.5	76.3	4.7	3.5	13.1	78.8	10.3	80.8	7.7	82.8	4.3	85.3	(4)
(5)			WB	14.8	66.9	4.6	1.9	11.5	68.3	8.8	69.4	6.3	70.5	3.0	71.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	46.8	35.8	35.0	37.6	43.2	49.4	54.7	59.5	60.3	56.1	48.9	42.4	37.8		
	DBStd	10.30	6.20	6.22	4.77	4.54	5.14	4.36	3.98	3.80	3.79	4.75	5.68	6.93		
	HDD50	2222	441	420	386	210	71	6	0	0	3	76	230	379		
	HDD65	6667	906	840	851	653	485	313	178	154	268	498	677	844		
	CDD50	1050	0	0	0	7	52	146	294	320	186	42	3	0		
	CDD65	20	0	0	0	0	1	2	7	9	1	0	0	0		
	CDH74	44	0	0	0	0	1	3	28	12	0	0	0	0		
	CDH80	1	0	0	0	0	0	0	1	0	0	0	0	0		
	WSAvg	14.5	16.7	14.4	14.1	12.8	13.1	13.7	13.1	13.5	14.9	15.8	15.7	16.0		
	PrecAvg	65.5	7.6	5.5	4.7	3.7	3.0	3.3	3.9	5.5	6.1	7.5	7.2	7.2		
(15) Precipitation	PrecMax	84.1	15.0	11.9	10.4	5.9	6.9	7.6	10.0	9.3	11.7	14.6	14.1	13.4		
	PrecMin	40.5	0.4	0.7	0.9	1.7	0.6	0.3	0.6	1.0	2.0	2.0	3.3	0.9		
	PrecStd	11.3	3.2	3.0	2.2	1.2	1.6	1.8	2.0	1.9	2.4	3.2	3.1	3.4		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	47.7	45.7	51.6	61.2	70.7	73.2	78.3	75.6	69.1	60.0	54.3	50.1		
		MCWB	45.7	43.8	43.6	50.4	57.7	60.0	65.1	64.1	62.7	55.5	52.8	49.0		
	2%	DB	46.0	44.5	46.3	56.3	65.3	68.0	72.3	72.1	65.3	57.9	52.8	48.8		
		MCWB	44.2	43.0	41.7	48.4	55.7	59.6	63.1	63.7	59.9	54.7	51.0	47.4		
	5%	DB	44.9	43.7	44.3	52.9	61.0	64.8	68.6	68.9	63.2	56.5	51.3	47.7		
		MCWB	43.2	42.3	41.7	46.2	53.8	58.2	62.2	62.2	58.7	53.7	49.2	46.1		
	10%	DB	43.9	42.8	43.0	49.9	57.3	61.6	66.0	66.6	61.6	55.4	50.0	46.6		
		MCWB	42.3	41.6	41.0	45.2	51.8	56.6	61.1	61.2	57.8	52.9	47.9	45.0		
	0.4%	WB	46.4	44.8	45.1	52.5	60.8	62.9	67.3	66.8	64.0	57.2	53.7	49.4		
		MCDB	47.4	45.2	48.2	57.6	67.9	68.0	74.5	72.1	67.0	58.8	54.0	49.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	44.7	43.7	43.5	49.4	57.0	61.0	64.9	64.8	61.5	55.7	51.3	47.8		
		MCDB	45.5	44.2	45.3	54.2	63.3	66.0	70.5	69.9	64.0	57.0	52.3	48.3		
	5%	WB	43.7	42.7	42.4	47.5	54.5	59.1	63.1	63.3	59.8	54.5	49.7	46.6		
		MCDB	44.7	43.3	43.8	51.5	59.9	63.3	67.5	67.7	62.3	56.0	50.9	47.4		
	10%	WB	42.6	41.8	41.4	45.8	52.3	57.2	61.5	61.9	58.4	53.3	48.2	45.2		
		MCDB	43.8	42.7	42.7	49.0	56.5	60.9	65.3	65.6	61.0	55.0	49.7	46.4		
	5% DB	MDBR	6.1	6.5	6.7	8.3	8.3	7.7	7.7	8.2	7.5	7.0	6.5	6.6		
		MCDBR	5.9	5.5	7.6	13.1	14.4	13.6	13.3	11.9	10.0	7.6	6.4	5.5		
		MCWBR	6.3	5.1	5.9	8.2	8.9	8.6	8.0	7.4	6.9	6.3	6.9	6.1		
		MCWBR	5.8	4.6	6.2	11.0	13.0	11.4	11.2	10.1	8.6	6.4	6.3	5.2		
(35) Mean Daily Temperature Range	5% WB	MCWBR	6.3	4.8	5.2	7.3	8.6	8.2	7.6	7.2	7.0	6.2	7.1	6.1		
	taub		0.315	0.337	0.354	0.387	0.374	0.365	0.379	0.384	0.363	0.345	0.319	0.293		
		taud	2.190	2.276	2.326	2.294	2.404	2.455	2.435	2.418	2.463	2.397	2.264	2.158		
(40) Clear-Sky Solar Irradiance	Ebn at Noon		160	217	250	260	273	277	270	259	247	214	159	129		
		Edh at Noon	16	23	29	35	34	32	32	31	25	21	15	12		
	RadAvg		108	300	727	1217	1641	1840	1658	1266	821	393	149	74		
		RadStd	22	61	78	136	197	169	180	103	85	61	31	10		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (26 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	-220	N/A	N/A

Nomenclature: See separate page